

TEST ON TILES & MARBLES

Here are the **common tests on Tiles and Marbles used in construction** for quality checking:

Tests on Tiles

1. **Water Absorption Test**
2. **Breaking Strength Test**
3. **Abrasion Resistance Test**
4. **Impact Resistance Test**
5. **Hardness Test**
6. **Chemical Resistance Test**
7. **Dimensional Accuracy Test**
8. **Surface Quality Test**

Tests on Marble

1. **Water Absorption Test**
2. **Compressive Strength Test**
3. **Flexural Strength Test**
4. **Abrasion Resistance Test**
5. **Impact Test**
6. **Hardness Test**
7. **Chemical Resistance Test**

Tests on Tiles

Water Absorption Test



Breaking Strength Test



Hardness Test



Chemical Resistance Test



Tests on Marbles

Abrasion Resistance Test



Impact Resistance Test



Dimensional Accuracy Test



Surface Quality Test



Tests on Marbles

Water Absorption Test



Compressive Strength Test



Dimensional Accuracy Test



Surface Quality Test



• Tiles • Mohs Hardness picks

Tests on Marbles

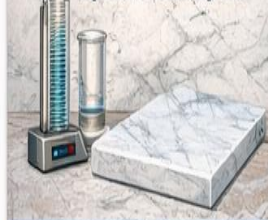
Water Absorption Test



Chemical Resistance Test



Specific Gravity Test



Chemical Resistance Test



• Marble

Water Absorption Test on Tiles and Marbles

Purpose:

The water absorption test is used to determine the **amount of water absorbed by tiles and marble**. It helps to check the **porosity, durability, and quality** of the material.

Apparatus

- Weighing balance
- Water container / tray
- Oven (105–110°C)
- Dry cloth

Procedure

1. **Dry the specimen** (tile or marble) in an oven at **105–110°C** for about **24 hours**.
2. Allow the specimen to **cool at room temperature**.
3. **Weigh the dry specimen** and record it as **W_1 (Dry Weight)**.
4. Immerse the specimen in **clean water for 24 hours**.
5. Remove it from water and **wipe the surface with a cloth**.
6. **Weigh the wet specimen** and record it as **W_2 (Wet Weight)**.

Formula

$$\text{Water Absorption (\%)} = \frac{W_2 - W_1}{W_1} \times 100$$

Where:

- **W_1** = Dry weight of specimen
- **W_2** = Wet weight after immersion

Result / Standard Values

- **Tiles:** Water absorption generally $\leq 3\text{--}10\%$ depending on type.
- **Marble:** Water absorption usually $\leq 0.4\%$.

Tests on Tiles

Water Absorption Test



Breaking Strength Test



Hardness Test



Chemical Resistance Test



Tests on Marbles

Abrasion Resistance Test



Impact Resistance Test



Dimensional Accuracy Test



Surface Quality Test



Tests on Marbles

Water Absorption Test



Compressive Strength Test



Dimensional Accuracy Test



Surface Quality Test



Tests on Marbles

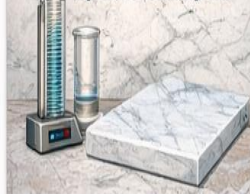
Water Absorption Test



Chemical Resistance Test



Specific Gravity Test



Chemical Resistance Test



• Tiles • Mohs Hardness picks

• Marble

Breaking Resistance Test on Tiles and Marbles

Purpose:

The breaking resistance test is used to determine the **maximum load a tile or marble slab can withstand before breaking**. It helps evaluate the **strength and durability** of the material for flooring and wall applications.

Apparatus

- Universal Testing Machine (UTM) or Breaking Strength Testing Machine
- Two support rollers
- Loading roller
- Measuring scale

Procedure

1. Place the **tile or marble specimen on two support rollers** of the testing machine.
2. Ensure the **tile is properly aligned and supported**.
3. Apply **load gradually at the center of the specimen** using the loading roller.
4. Increase the load until the **tile or marble breaks**.
5. Record the **maximum load at failure**.

Result

The **breaking resistance** is expressed as the **maximum load (in Newtons or kg) the specimen can withstand before breaking**.

BREAKING STRENGTH TEST OF TILES & MARBLES



Abrasion Resistance Test on Tiles and Marbles

Purpose:

The abrasion resistance test is used to determine the **resistance of tiles and marble surfaces to wear and friction** caused by foot traffic. This test helps evaluate the **durability and surface hardness** of flooring materials.

Apparatus

- **Abrasion testing machine** (Tile Abrasion Tester / Los Angeles Abrasion Machine)
- Abrasive powder or abrasive disc
- Weighing balance
- Measuring scale

Procedure

1. **Clean and dry the tile or marble specimen** properly.
2. **Measure and record the initial weight or thickness** of the specimen.
3. Place the specimen on the **abrasion testing machine**.
4. Allow the machine to **rotate with abrasive powder or disc** for a specified number of cycles.
5. After the test, **remove the specimen and clean it**.
6. **Measure the final weight or thickness** of the specimen.
7. Calculate the **loss in weight or thickness due to abrasion**.

Result

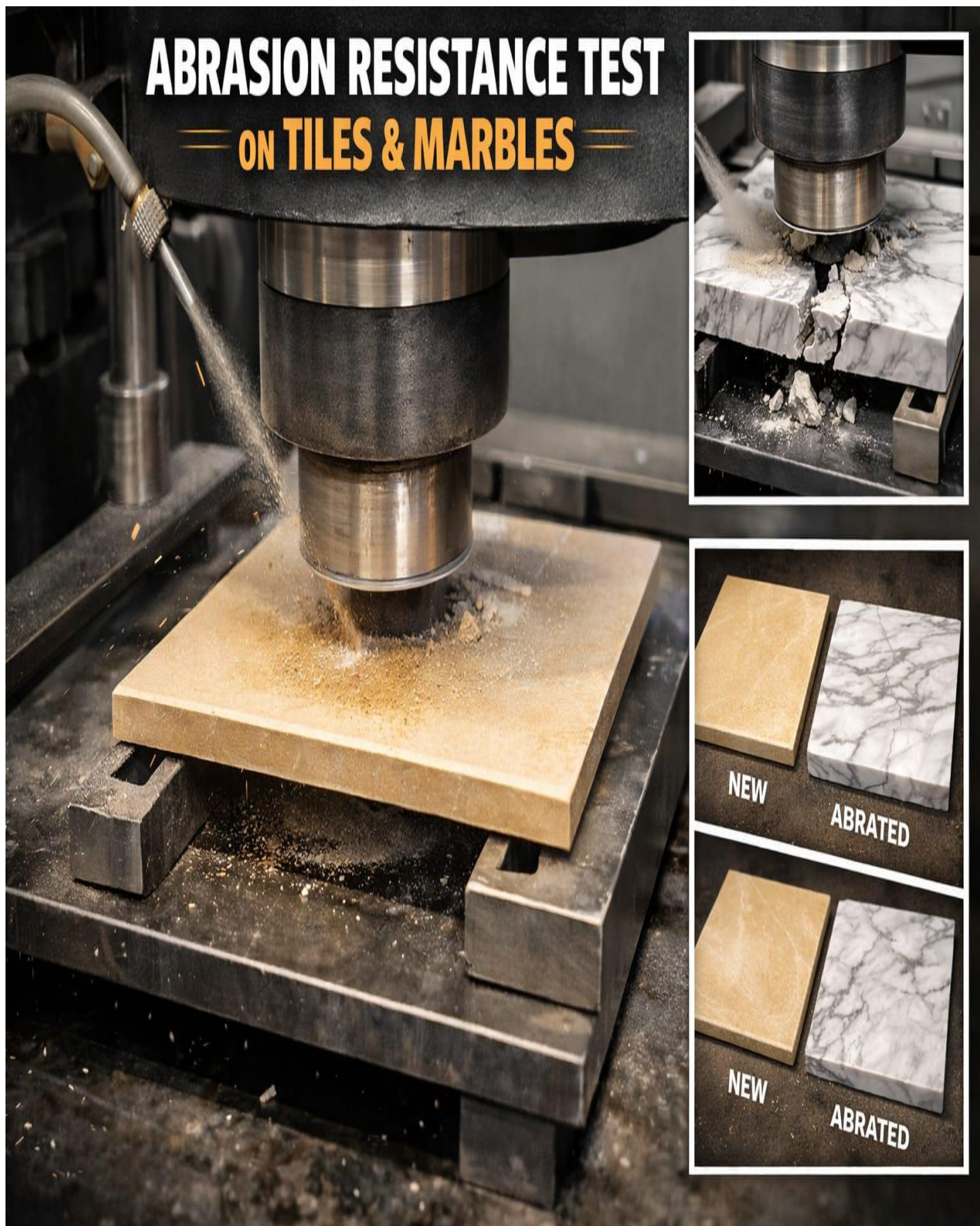
Abrasion resistance is expressed as:

Loss in weight (g) or loss in thickness (mm) after the abrasion test.

Lower loss = **Higher abrasion resistance and better durability**.

ABRASION RESISTANCE TEST

ON TILES & MARBLES



Impact Resistance Test on Tiles and Marbles

Purpose:

The impact resistance test is used to determine the **ability of tiles and marble to withstand sudden shocks or impacts** without cracking or breaking.

Apparatus

- Impact testing machine or **steel ball / drop weight apparatus**
- Supporting frame or base
- Measuring scale

Procedure

1. Place the **tile or marble specimen on a flat and rigid support**.
2. Position the **steel ball or weight above the specimen**.
3. Drop the weight from a **specified height** onto the center of the specimen.
4. Observe whether the **tile or marble cracks or breaks**.
5. Repeat the test by **increasing the height** if required until failure occurs.

Result

The **impact resistance** is determined by the **maximum height or energy at which the tile or marble can withstand the impact without breaking**.

Importance

- Determines the **toughness and durability** of tiles and marble.
- Ensures suitability for **flooring in heavy traffic areas**.
- Helps prevent **cracking due to accidental impacts**.

Hardness Resistance Test on Tiles and Marbles

Purpose:

The hardness resistance test is used to determine the **ability of tiles and marble surfaces to resist scratches and wear**. It helps evaluate the **surface strength and durability** of the material.

Apparatus

- **Mohs Hardness Testing Kit** (hardness picks)
- Steel needle or scratching tool
- Magnifying glass

Procedure

1. **Clean the surface** of the tile or marble specimen.
2. Select a **Mohs hardness pick** with a known hardness value.
3. **Scratch the surface lightly** with the hardness pick.
4. Observe whether a **visible scratch is formed** on the surface.
5. Repeat with picks of **different hardness levels** until a scratch appears.

Result

- The **hardness of the tile or marble** is equal to the **Mohs hardness number of the pick that just scratches the surface**.

Example:

- If the surface scratches with **Mohs hardness 6**, the hardness of the material is **about 6**.

Importance

- Determines **scratch resistance** of tiles and marble.
- Helps select materials for **flooring and high-traffic areas**.
- Indicates **surface durability and quality**.

Hardness Test on Tiles and Marbles

1 Select a Mohs Hardness Pick

2 Scratch Surface of Tile and Marble

3 Determine Hardness of Materials

Select a Mohs Hardness Pick

Scratched at Mohs #6

Scratched at Mohs #3

1	2	3	4	5
				
Talc	Gypsum	Calcite	Fluorite	Apatite

Mohs Scale of Hardness

 Water Absorption (%) = $\frac{W_2 - W_1}{W_1} \times 100$

W_1 = Dry Weight W_2 = Wet Weight

Chemical Resistance Test (Tiles and Marbles)

Purpose:

The chemical resistance test is used to determine the **ability of tiles and marble surfaces to resist damage from chemicals** such as acids, alkalis, and cleaning agents.

Apparatus

- Chemical solutions (acid, alkali, detergent)
- Dropper or pipette
- Water
- Soft cloth
- Magnifying glass

Procedure

1. **Clean the surface** of the tile or marble specimen.
2. Place a few **drops of chemical solution** (acid, alkali, or detergent) on the surface.
3. Leave the chemical on the surface for **about 24 hours**.
4. Wash the specimen with **clean water** and dry it with a cloth.
5. **Inspect the surface** for any damage such as **stains, discoloration, or surface erosion**.

Result

- If **no visible change** occurs → Good chemical resistance.
- If **stains, dullness, or erosion** appear → Poor chemical resistance.

Importance

- Ensures tiles and marble can **resist cleaning chemicals**.
- Prevents **surface damage and discoloration**.
- Important for **kitchens, bathrooms, and laboratories**.

Chemical Resistance Test on Tiles and Marbles

1 Apply Chemicals



- Place drops of acids, alkalis, and cleaners on the surface



Select Chemicals

2 Allow to Sit for 24 Hours



- Let the chemicals sit for 24 hours

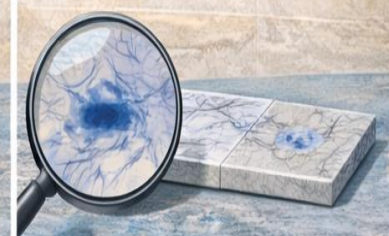


Strained #4 at Mohs #6

3 Clean the Surface



- Examine for damage or discoloration



Strained at Mohs #6

1



Tale

2



Gypum

3



Calcite

4



Fluorite

5



Marite

6



Topaz

8



Topaz

10



Mescale

Results

- **Tile** anter-Erozed surface
- **Marble:** Derkred surface

Surface Quality Test on Tiles and Marbles

Purpose:

The surface quality test is used to check the **appearance and finish of tiles and marble**. It ensures that the surface is **smooth, free from defects, and suitable for construction use**.

Apparatus

- Straight edge
- Magnifying glass
- Measuring scale
- Good lighting

Procedure

1. Place the **tile or marble on a flat surface**.
2. **Visually inspect the surface** under good lighting conditions.
3. Use a **magnifying glass** to check for small defects.
4. Check the surface using a **straight edge** to detect warping or unevenness.
5. Measure the **length, width, and thickness** to verify uniformity.

Defects to Check

- **Cracks**
- **Chipped edges**
- **Scratches**
- **Surface stains or discoloration**
- **Warping or uneven surface**

Result

- If the surface is **smooth and free from defects**, the tile or marble is **acceptable**.
- If **cracks, chips, or uneven surfaces** are found, the material is **rejected or classified as lower grade**.

Importance

- Ensures **good appearance and finishing quality**.
- Prevents **defective materials from being used in construction**.
- Improves **durability and aesthetics of flooring and wall finishes**.

Surface Test on Tiles and Marbles

1 Visual Inspection	2 Straight Edge Test	3 Thickness Measurement	4 Dimension Measurement
			
• Check tiles and marble visually under good light for surface defects	• Use a straight edge and filler gauge to check flatness and warping	• Measure tile/marble thickness with digital caliper	• Check tile/marble length and width for dimensional accuracy

Tools: Flashlight • Magnifying Glass • Flashlight pen

Check for:

- Chipped edges
- Cracks
- Scratches
- Warping

Results

			
Chipped Edge	Cracked Surface	Scratches	Warping

DIMENSIONAL ACCURACY TEST ON TILES AND MARLBES

Purpose:

The dimensional accuracy test is conducted to check whether **tiles and marble slabs have correct and uniform dimensions** such as length, width, and thickness according to the specified standards.

Apparatus

- Vernier caliper or digital caliper
- Measuring scale or steel ruler
- Straight edge

Procedure

1. Select a **sample of tiles or marble slabs** for testing.
2. Measure the **length and width** of the specimen using a **steel ruler or caliper**.
3. Measure the **thickness** of the specimen at different points.
4. Compare the measured dimensions with the **standard specified dimensions**.
5. Check the **edges and corners** to ensure they are straight and square.

Result

- If the measured dimensions are **within the permissible tolerance limits**, the tiles or marble are **dimensionally accurate**.
- If the difference is **greater than the allowed tolerance**, the material is **rejected or classified as defective**.

Importance

- Ensures **uniform size of tiles and marble slabs**.
- Helps in **proper alignment during installation**.
- Reduces **gaps and uneven surfaces in flooring and wall finishes**.

Flexural Strength Test of Tiles and Marbles

Purpose:

The flexural strength test is used to determine the **bending strength of tiles and marble slabs**. It shows the ability of the material to **resist bending or breaking under load**.

Apparatus

- **Universal Testing Machine (UTM)** or Flexural Testing Machine
- Two support rollers
- One loading roller
- Measuring scale

Procedure

1. Place the **tile or marble specimen horizontally on two support rollers**.
2. Ensure the specimen is **properly aligned and supported**.
3. Apply the **load gradually at the center** of the specimen using the loading roller.
4. Continue increasing the load until the **specimen breaks or cracks**.
5. Record the **maximum load at failure**.

Result

The flexural strength is expressed in **N/mm² or MPa**.

Importance

- Determines the **bending resistance of tiles and marble slabs**.
- Ensures **durability and strength for flooring and wall applications**.
- Helps select materials for **load-bearing surfaces**.